Texture-Aware Causal Feature Extraction Network for Multimodal Remote Sensing Data Classification

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***Abstract*— The pixel-level classification of multimodal remote sensing (RS) images plays a crucial role in the intelligent interpretation of RS data. However, existing methods that mainly focus on feature interaction and fusion fail to address the challenges posed by confounders—brought by sensor imaging bias, limiting their performance. In this article, we introduce causal inference into intelligent interpretation of RS and propose a new texture-aware causal feature extraction network (TeACFNet) for pixel-level fusion classification. Specifically, we propose a two-stage causal feature extraction (CFE) framework that helps networks learn more explicit class representations by capturing the causal relationships between multimodal heterogeneous data. In addition, to solve the problem of low-resolution land cover feature representation in RS images, we propose the refined statistical texture extraction (ReSTE) module. This module integrates the semantics of statistical textures in shallow feature maps through feature refinement, quantization, and encoding. Extensive experiments on two publicly available datasets with different modalities, including Houston2013 and Berlin datasets, demonstrate the remarkable effectiveness of our proposed method, which reaches a new state-of-the-art.**

***Index Terms*— Causal feature extraction, feature fusion, image pixel-level classification, multimodal remote sensing (RS), texture representation learning.**

I. Introduction

REMOTE sensing (RS) land cover classification plays a crucial role in the intelligent interpretation of RS, with the objective of assigning pixel-level labels in an image. Recent advancements in deep learning technology have significantly contributed to the growing application of land cover classification in various domains, including autonomous driving [1], building extraction [2], scene classification [3], [4], and even military. As the accessibility of diverse RS sensors continues to grow, it is now more important than ever to effectively utilize multimodal RS data to improve the accuracy of downstream tasks [5], [6], [7], [8], [9], [10].

In order to effectively capture and utilize multimodal information, a variety of feature interaction and fusion methods have been applied to RS images, which promotes the rapid development of pixel-level classification in this domain. For example, Wang et al. [11] and Hang et al. [12] explored weight-sharing strategies to enhance global feature interaction. Shen et al. [13] proposed a shuffle feature enhancement module to prompt self-interaction and mutual-interaction. FusAtNet [14] proposed a dual attention-based feature fusion and extraction framework. Multiscale feature extractor [15] and encoding–decoding structure [16], [17] are also relatively common techniques. In addition, the Transformer network architecture [18], primarily composed of attention modules, has experienced rapid advancements in computer vision. Roy et al. [19] successfully applied the Transformer in the context of multimodal RS data and achieved notable performance increases by fully capturing intermodal interactions. ExViT [20] showed good modality-agnostic expandability, which can adapt to data of different modalities. Hong et al. [21] proposed a novel cross-city multimodal semantic segmentation network. These feature interaction and fusion approaches have demonstrated their prowess in modeling long-term dependencies and learning global spatial representations of RS images. However, these existing models often overlook the impact of confounder, i.e., sensor imaging bias, on the multimodal data fusion process.

Sensor imaging bias denotes systematic differences or discrepancies in the captured data arising from the use of diverse sensors in RS [22]. Different sensors, such as hyperspectral (HS) sensors [23], optical sensors [24], or synthetic aperture radar (SAR) sensors [25], are often employed to capture diverse aspects of the same scene. However, these sensors may have inherent variations in their data acquisition processes, leading to biases in the captured data. When various modalities are exploited, these biases can lead to changes in the context distribution, hindering the accurate integration of information from different sources. Thus, solving sensor imaging bias is crucial for achieving reliable and robust intelligent interpretation of multimodal RS data. Methods as previously described rely heavily on the independent and identical distribution (IID) assumption for dataset. This hypothesis contradicts the imaging bias of multimodalities, which has been neglected in previous studies [26].

Motivated by causal inference, causal-informed neural networks have demonstrated the ability to solve dataset bias or imbalance problems in multiple domains and tasks, such as long-tail distribution [27], person re-identification [28], and domain generalization [29]. Li et al. [30] removed the effect of clothing on identity recognition by modeling

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causal interventions. Rao et al. [31] proposed a counterfactual attention learning method to improve visual attention learning. To the best of our knowledge, such approaches have been explored in various fields, while their application in RS land cover classification remains unexplored.

In this article, we address this research gap by integrating invariant risk minimization (IRM) [32], a causal-inspired algorithm, for the first time into RS land cover classification and propose a novel texture-aware causal feature extraction network (TeACFNet) for pixel-level fusion classification. Specifically, we propose a two-stage causal feature extraction (CFE) framework. First, we treat categories as a changing environment and apply IRM to extract class-equivariant features (ClEFs) that are consistent among members of different categories. It is accomplished by utilizing IRM on the intraclass sample similarity contrastive loss, which ensures a consistent level of similarity among intraclass samples across all classes. Apparently, the features are irrelevant to the classification labels, so they are named spurious features. Then, we fix the ClEF extractor and use inverse probability weighting (IPW) [33] to remove spurious features from the total features, leaving us with causal features. In addition, a refined statistical texture feature extraction module is introduced in the extractor to extract the texture information in the shallow feature maps.

The main contributions of this article can be summarized as follows.

1) We introduce IRM into multimodal RS land cover classification for the first time and propose a novel two-stage CFE framework to learn more explicit causal representation, thereby improving the model's generalization ability.
2) To extract features that are causally related to classification labels, we propose a novel method consisting of two key components: ClEF extraction and invariant features extractor with IPW (I^2FE). This two-stage methodology effectively addresses the challenge of context distribution changes arising from multimodal data, all accomplished without the necessity for additional attribute annotations.
3) The refined statistical texture extraction (ReSTE) module is proposed to effectively capture statistical texture details by analyzing local features and their interconnections within shallow feature maps, which significantly enhances the discriminative capabilities of the feature extractor.

The remainder of this article is organized as follows. In Section II, we review related works in the field of causal inference and texture representation learning. Section III presents the detailed description of the proposed TeACFNet. For the experimental analysis, including datasets, settings, evaluation metrics, and discussions, we refer to Section IV. Finally, we summarize our conclusions in Section V.

II. Related Work

A. Causal Inference

Recently, there has been increasing attention on improving deep learning with causal inference [34]. Causal inference offers a robust ability to mitigate bias and focus on understanding causal effects [35], [36], making it applicable to a wide range of computer vision tasks, such as semantic segmentation [37], [38], visual question answering (VQA) [39], video inpainting [40], anomaly detection [41], and scene graph generation [42]. However, prior to the introduction of IRM, the utilization of causality in deep learning was limited. Because many methods require stronger assumptions, such as knowing the entire causal graph structure [43] or the specific intervention location and type [30], which often need additional labels for confounders. In contrast, IRM offers several advantages [32]. First, it only requires collecting a diverse range of environments with interventions, without the need for precise knowledge about the intervention form or location. Second, IRM aims to extract causal features that are invariant to the label, focusing only on the immediate parent node of the label. Therefore, assumptions about other parts of the causal graph, such as spurious features, are not necessary. Based on these advantages, in this article, we creatively regard the category label as an environment and propose a novel two-stage CFE framework.

B. Texture Representation Learning

Texture representation learning has emerged as a prominent area of research in the domain of visual perception and image processing. Recent advancements in deep learning have shifted the focus of texture representation toward explicit material identification through effective texture encoding techniques [44], [45], [46]. These methods aim to learn essential low-level features that are crucial for material classification and semantic segmentation. To enhance the effectiveness of low-level features, some approaches incorporate skip connections [47] or concatenation of texture-related information [48], [49]. In addition, the method in [50] exploited boundary information for feature propagation or supervised object edges. However, the focus of these techniques mainly revolves around structural features [51], and they are not designed to account for the low resolution typically observed in RS images. In contrast, our proposed method places additional emphasis on statistical texture features and introduces a targeted refinement network specifically designed for RS images.

III. Methodology

A. Problem Definition

In supervised learning of multimodal RS data, the generalization of the model relies on the fundamental assumption that any patch x can be generated from disentangled features, i.e., $x = f(x_c, x_s)$, where x_c represents the causal features, x_s represents the spurious features (e.g., ClEFs), and $f(\cdot)$ is a generator function that maps these features from vector space to sample space (e.g., pixels). Causal features are the direct causes of the outcome, while spurious features are confounders that exhibit correlations with the labels but lack a causal link [52]. Furthermore, if we use Y to represent the category label, then $P(Y|x_s)$ is unstable (this conditional distribution can be simply understood as the correlation between the two),

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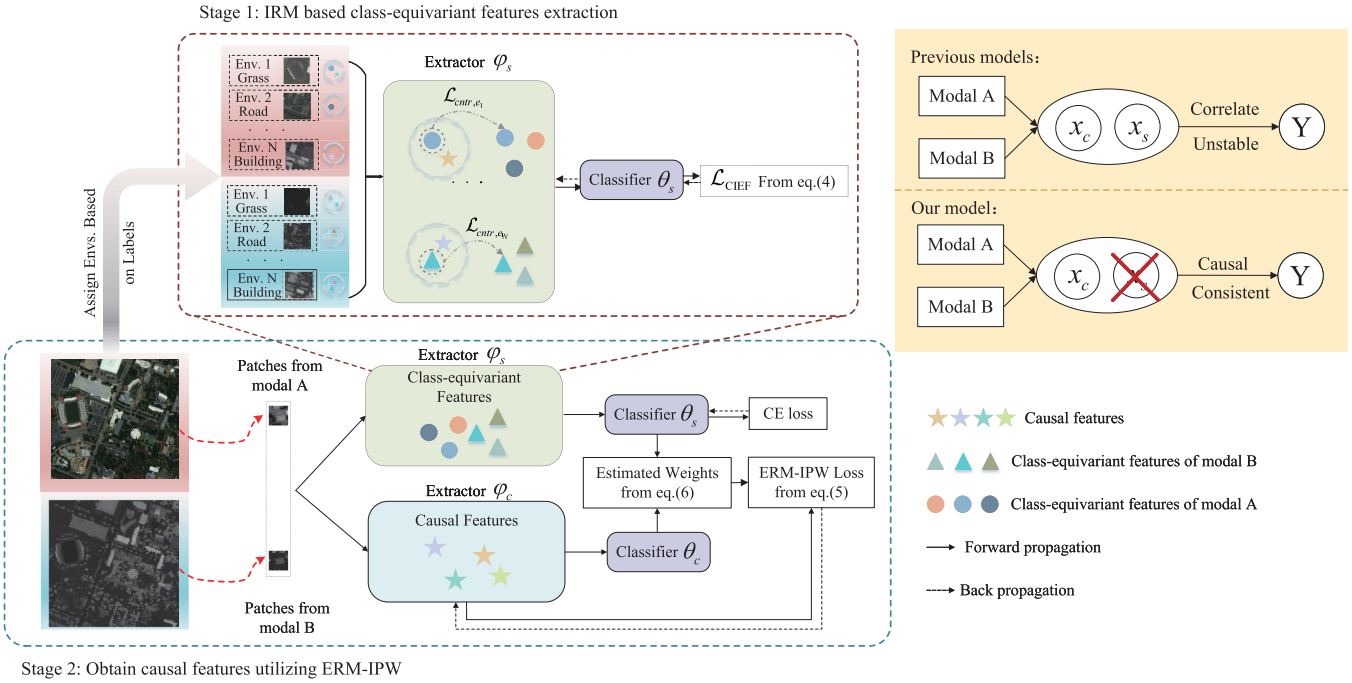


Fig. 1. Overall architecture of the proposed TeACFNet. TeACFNet introduces a two-stage training regimen with a shared backbone structure detailed in Fig. 2. In the first stage, the classifier θ_s remains fixed, allowing the extractor φ_s to focus on identifying ClEFs. These features are consistent across different categories within multimodal datasets. In the second stage, the extractor φ_s, now trained, is kept constant to isolate ClEFs. Utilizing IPW, these features (x_s) are filtered from the overall feature maps extracted by, thus extracting invariant features (x_c).

and $P(Y|x_c)$ is stable. Our goal is to learn a machine learning model that depends only on x_c.

If the supervised training dataset is perfectly balanced, i.e., the samples of each class are as diverse as possible, and the distribution of samples among all classes is as uniform as possible, then the widely used softmax cross-entropy (CE) loss can calmly preserve causal features by penalizing interclass similarity and remove spurious features by favoring intraclass similarity. However, for multimodal datasets, detectors with different imaging mechanisms have inherent characteristics, and the distribution shift they bring is unlabeled, introducing strong data imbalance issues that challenge the direct application of empirical risk minimization (ERM) [26], [53].

Consequently, we aim to learn a feature extractor $\varphi_c(x) = x_c$ that can extract causal features of the data, enabling the subsequent classifier θ to make robust predictions on multimodal heterogeneous data.

B. Overview

To enhance the performance of multimodal RS image classification, we propose TeACFNet. It consists of three parts: ClEF extraction, I^{2}FE, and ReSTE module.

Our proposed TeACFNet integrates the principles of IRM and texture representation learning. The overall framework of TeACFNet is illustrated in Fig. 1. The detailed backbone architecture is shown in Fig. 2. First, our approach utilizes a two-stage CFE framework, consisting of ClEF and I^{2}FE. By treating labeled classes as changing environments, we effectively address the context distribution changes introduced by multimodal data. Second, we enhance the commonly used forward-inverted CNN [54] by adding a novel

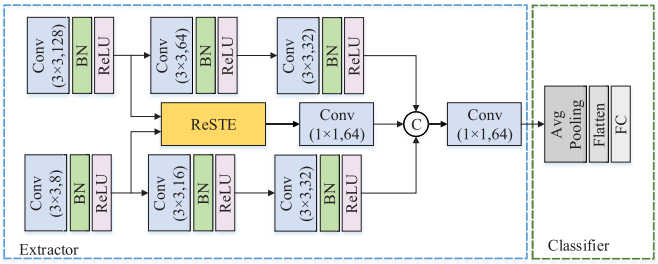


Fig. 2. Detailed backbone architecture of our proposed TeACFNet, including the extractor and classifier.

ReSTE module, as depicted in Fig. 3. This module is capable of extracting refined texture information from multimodal RS images. By leveraging statistical texture analysis techniques, ReSTE facilitates the identification of patterns, structures, and pixel intensity variations within local image regions. This empowers our model to distinguish complex land cover classes, especially in regions with different spectral or spatial signatures but the same labels.

C. ClEF Extraction

In the real world, the data we collect are often imperfectly balanced, making it a tough challenge to learn without bias. This imbalance in datasets can significantly hinder our ability to recognize consistent patterns, i.e., causal features, across different categories. The IRM algorithm provides a promising solution. If we consider each class as an environment, IRM enables the extraction of ClEFs that are consistent among members of different categories. If we can remove these spurious features in the total feature maps, then the derived feature

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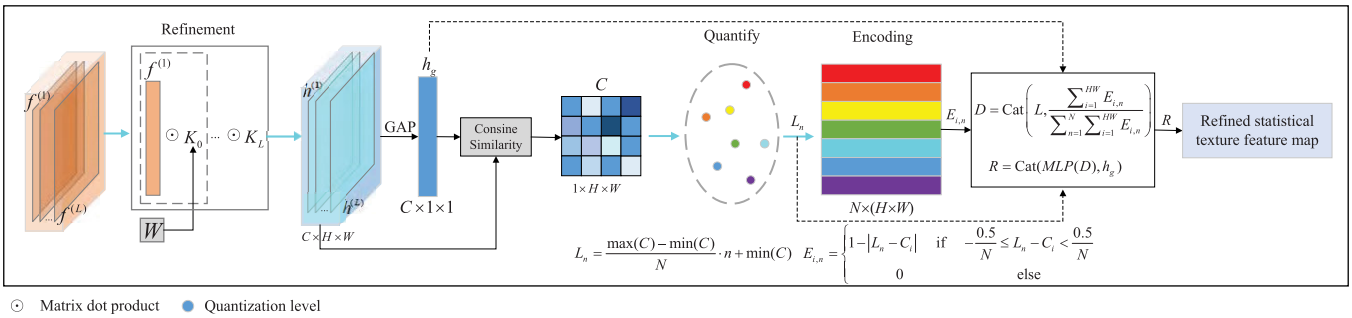


Fig. 3. Illustration of the ReSTE module.

representation becomes more robust and discriminative, as it focuses solely on the essential characteristics of the class, disregarding spurious confounders introduced by the different imaging modalities.

Unlike ERM, which achieves invariance in a passive manner through random experiments on balanced data, IRM provides a powerful framework for learning invariance by actively intervening in the learning process [32]

$$\min_{\varphi,\theta} \quad \sum_{e} R^e(\theta \cdot \Phi)$$
$$\text{subject to } \theta \in \underset{\bar{\theta}}{\operatorname{argmin}} R^e\left(\bar{\theta} \cdot \Phi\right) \quad \forall e \in \mathcal{E} \tag{1}$$

where $\mathcal{E}$ is the set of environments, $R^e(\theta \cdot \Phi)$ represents the empirical risk specifically within the environment e, $\Phi = \varphi(x)$ is the representions of input images (φ is the feature extractor), and θ is the classifier.

Since (1) is a two-level optimization problem, it can be simplified to a single-variable optimization problem as follows:

$$\min_{\varphi} \sum_{e} R^e(\Phi) + \lambda \left\| \nabla_\theta R^e(\theta \cdot \Phi) \right\|^2 \tag{2}$$

where $\lambda > 0$ is the invariance regularization term.

To effectively implement IRM, we adopt an active intervention approach by treating different class labels as environments and replace $R^e(\theta \cdot \Phi)$ with a contrastive loss, $\mathcal{L}_{\text{cntr}}$, that utilizes the intraclass sample similarity

$$\mathcal{L}_{\text{cntr}}(\varphi_t, e, \theta) = \sum_{x_i \in e} -\log \frac{\exp\left(\varphi_t(x_i)^T \varphi_t(\text{Aug}(x_i)) \cdot \theta\right)}{\sum_{x_i' \in e} \exp\left(\varphi_t(x_i)^T \varphi_t\left(x_i'\right) \cdot \theta\right)} \tag{3}$$

where $\text{Aug}(\cdot)$ represents common augmentations, such as rotations and flips, typically used in standard contrastive losses [55]. The adoption of contrastive loss is motivated by its ability to retain all intrinsic features present in each sample. Then, based on (2), our proposed ClEF loss is

$$\mathcal{L}_{\text{ClEF}}(\varphi_s, \theta) = \sum_{e} \frac{1}{|e|} \left[\mathcal{L}_{\text{cntr}}(\varphi_s, e, \theta) + \lambda \left| \nabla_\theta \mathcal{L}_{\text{cntr}}(\varphi_s, e, \theta) \right| \right]. \tag{4}$$

This loss decouples the ClEFs, i.e., achieves $\varphi_s(x) = x_s$.

The obtained ClEFs serve as a crucial foundation for the I^{2}FE. These features capture the environment–equivariant characteristics specific to classes and provide valuable information about the underlying properties of the land cover classes in RS data.

D. Invariant Features Extractor With IPW

IPW is a statistical technique commonly used in causal inference [33]. We can utilize it to reweight the samples during the classification process. The weights are estimated based on the ClEFs. This reweighting process helps improve the overall classification performance, particularly in context distribution changes caused by diverse imaging mechanisms.

The approach begins with the ERM-IPW loss [33], which incorporates a sample-level IPW term, denoted as $P(x_i \mid \varphi_s(x_i))$

$$\mathcal{L}_{\text{ERM-IPW}}(\varphi_c, \varphi_s, \theta) = \frac{1}{N} \sum_{i=1}^{N} \text{CE}\left(y_i, \hat{y}_i\right) \cdot \frac{1}{P(x_i \mid \varphi_s(x_i))} \tag{5}$$

where CE represents the CE loss, x_i represents the ith input patch, y_i represents the real label of x_i, and $\hat{y}_i$ represents the predicted label of x_i. We can observe that $(1/P(x_i \mid \varphi_s(x_i)))$ is the main difference between ERM-IPW and ERM. The implication of IPW is that when the input patch x is more strongly associated with its corresponding ClEF maps x_s and, by extension, the corresponding causal feature map x_c is also more likely to be associated with x_s, we should assign a lower weight to the loss function. The purpose of this weighting is to discourage any biases introduced by the spurious features and ensure a more balanced representation.

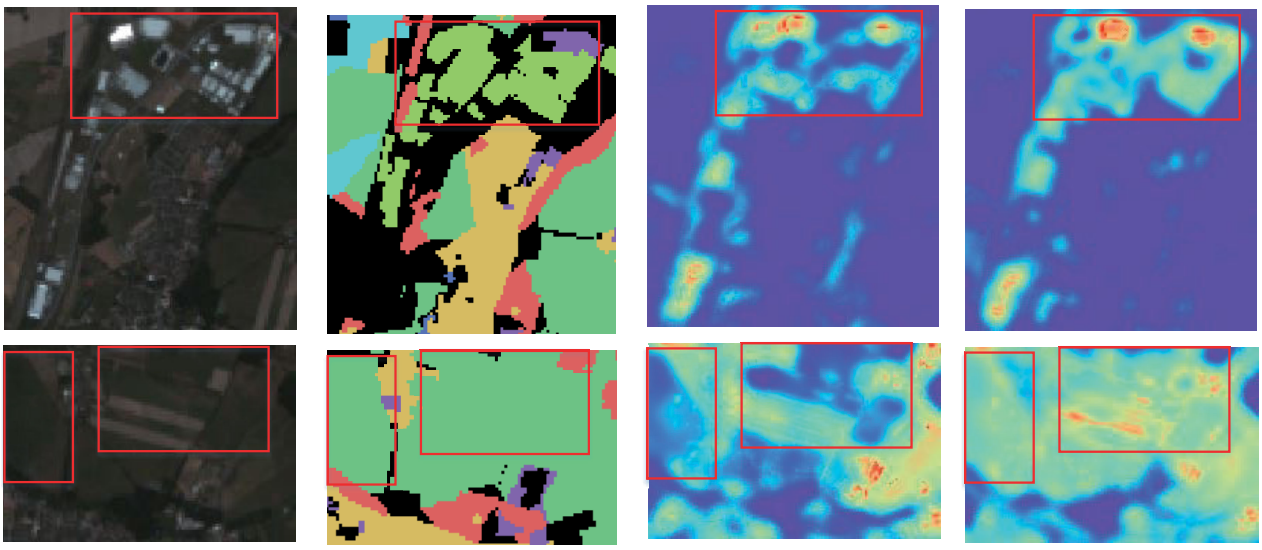
Thanks to ClEF, we have actually done the estimation of φ_s. Combined with the method to estimate spurious features using a biased classifier [56], [57], we can further improve IPW

$$P(x \mid \varphi_s(x)) \propto \frac{\text{CE}\left(y, \hat{y} = \theta_c(\varphi_c(x))\right) + \text{CE}\left(y, \hat{y} = \theta_s(x_s)\right)}{\text{CE}\left(y, \hat{y} = \theta_s(x_s)\right)} \tag{6}$$

where $x_s = \varphi_s(x)$.

Our experiments demonstrate that the proposed approach allows accurate extraction of invariant (causal) features with φ_c. Utilizing φ_s, we further enhance the IPW by estimating weights based on both the causal features and ClEFs. The biased classifier θ_s is trained using the CE loss. Our biased classifier provides more accurate weights for improved reweighting compared to traditional approaches.

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(a) (b) (c) (d)

Fig. 4. Qualitative results of our ReSTE module. It can be seen that pixels of the same category obtain similar feature activation intensity after applying the ReSTE module. Notice how the "industrial area" in the first row of (d) gets a significant increase in activation similarity across concrete building pixel locations, compared to the original feature output. Similarly, the second line has a similar effect. (a) Pseudo-color map. (b) GT. (c) Before ReSTE. (d) After ReSTE.

E. ReSTE Module

The ReSTE module effectively utilizes texture information for accurate pixel-level classification in low-resolution RS images. Traditional deep learning approaches often struggle to fully exploit texture information due to the lack of explicit modeling and integration of statistical texture cues. This limitation underscores the necessity of enhancing the utilization of texture representation during the classification process.

ReSTE aims to learn statistical texture in shallow feature maps and enhance classification accuracy by refining and incorporating statistical texture cues. The module comprises three key steps: feature refinement, quantization, and encoding. Fig. 3 illustrates the structure of ReSTE. In addition, Fig. 4 shows some samples of refined statistical texture features produced by applying ReSTE.

In the feature refinement step, ReSTE focuses on refining texture representation of shallow feature maps to increase their impact on deeper layers. This is crucial, especially in low-resolution or low-contrast images where texture details may be less visible. The feature refinement process enhances the texture features by considering the local features and their relationships, which is achieved by adjusting the weights of the convolutional kernel $\{K_0, \ldots, K_L\}$ based on a localized feature-constrained kernel [58].

In the feature quantization step, we first utilize global average pooling (GAP) on refined feature maps $\{h^{(1)}, \ldots, h^{(L)}\}$ to obtain the average feature map, denoted as h_g. Then, for each patch $h^l_{m,n} \in \{h^{(1)}, \ldots, h^{(L)}\}$, which is in channel l and is centered at (m, n), we calculate the cosine similarity between $h^l_{m,n}$ and h_g, resulting in a similarity map $C \in {}^{1\times H\times W}$. Next, we directly perform quantization and statistical analysis on reshaped $C \in {}^{1\times HW}$ to extract features similar to histograms. To achieve this, we first need to determine several quantization levels that can be considered as histogram bands. We employ a simple method by evenly dividing the range between the minimum and maximum values of C into N parts, resulting in N quantization levels. Specifically, the nth quantization level is defined as follows:

$$L_n = \frac{\max(C) - \min(C)}{N} \cdot n + \min(C). \tag{7}$$

In the feature encoding step, ReSTE generates a statistical feature map by encoding the global average feature and propagating information from all levels. Specifically, each pixel C_i ($i \in [1, \mathrm{HW}]$) is encoded into an N-dimensional vector E_i, n, and E is named the quantized encoding map

$$E_{i,n} = \begin{cases} 1 - |L_n - C_i|, & \text{if} -\dfrac{0.5}{N} \le L_n - C_i < \dfrac{0.5}{N} \\ 0, & \text{else.} \end{cases} \tag{8}$$

Subsequently, a 2-D map D that carries statistical significance is obtained as follows:

$$D = \mathrm{Cat}\left(L, \frac{\sum_{i=1}^{\mathrm{HW}} E_{i,n}}{\sum_{n=1}^{N}\sum_{i=1}^{\mathrm{HW}} E_{i,n}}\right). \tag{9}$$

D itself represents statistical information; however, it is important to note that, in order to preserve the high-dimensional information during the statistical analysis, the objects being analyzed are relative values (cosine similarity with the average feature). Therefore, to further obtain statistical information in absolute terms, we also encode the average feature into the analysis, obtaining R

$$R = \mathrm{Cat}\big(\mathrm{MLP}(D), h_g\big). \tag{10}$$

By combining these three steps, the network is able to capture important spatial and structural features conveyed by textures and incorporate it into the classification process, thus improving the richness of causal feature maps obtained by the extractor. The ReSTE module stands as a vital assurance for the effectiveness of our proposed two-stage CFE method, especially in the case of low-resolution RS images.

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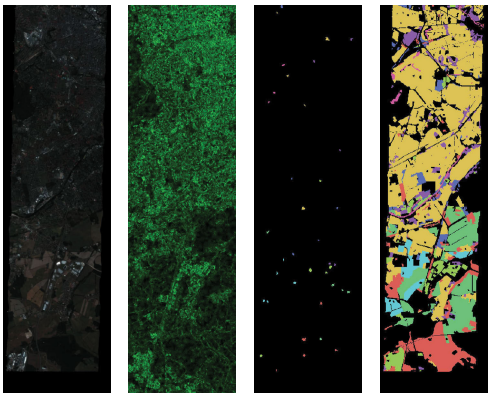


Fig. 5. Visual representation of Berlin dataset. (a) Pseudo-color image of HS data, showcasing bands 59, 40, and 23. (b) Pseudo-color image of SAR data, showcasing bands 2, 4, and 3. (c) GT maps for training. (d) GT maps for testing.

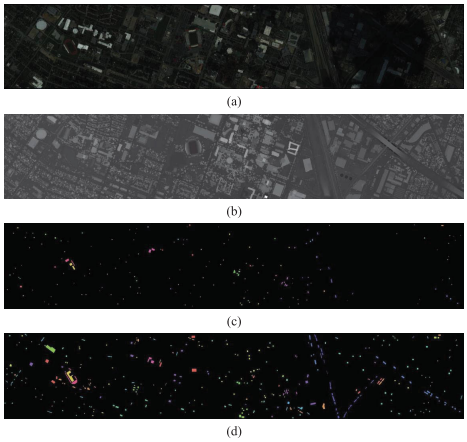


Fig. 6. Visual representation of Houston2013 dataset. (a) Pseudo-color image of HS data, showcasing bands 59, 40, and 23. (b) Grayscale image of LiDAR data. (c) GT maps for training. (d) GT maps for testing.

TABLE I
COMPREHENSIVE DESCRIPTION OF BERLIN DATASET

No	Class Name	Testing set	Training Set
1	Forest	54511	443
2	Residential Area	268219	423
3	Industrial Area	19067	499
4	Low Plants	58906	376
5	Soil	17095	331
6	Allotment	13025	280
7	Commercial Area	24526	298
8	Water	6502	170
-	Total	461851	2820

TABLE II
COMPREHENSIVE DESCRIPTION OF HOUSTON2013 DATASET

No	Class Name	Testing Set	Training Set
1	Healthy grass	1053	198
2	Stressed grass	1064	190
3	Synthetic grass	505	192
4	Tree	1056	188
5	Soil	1056	186
6	Water	143	182
7	Residential	1072	196
8	Commercial	1053	191
9	Road	1059	193
10	Highway	1036	191
11	Railway	1054	181
12	Parking lot 1	1041	192
13	Parking lot 2	285	184
14	Tennis court	247	181
15	Running track	473	187
-	Total	12,197	2832

IV. EXPERIMENTS

A. Datasets

1) HS-SAR Berlin Data: It captures both urban and rural areas of Berlin with eight land cover classes. It comprises two main data sources: 1) HS image derived from HyMap HS data with a spatial resolution of 30 m, which has a resolution of 797 × 220 pixels and contains 244 spectral bands ranging from 400 to 2500 nm, and 2) SAR data obtained from the Sentinel-1 dual-Pol (VV–VH) sensor covering the same geographical region, which comprises an image of 1723 × 476 pixels with a spatial resolution of 13.89 m. The ground-truth (GT) map was generated using OpenStreetMap data. Fig. 5 provides a visual representation and details on the training set and testing set can be found in Table I.

2) HS-LiDAR Houston2013 Data: It was captured by the ITRES CASI-1500 sensor and used in the 2013 IEEE GRSS Data Fusion contest. The dataset comprises two data sources: an HS image and a light detection and ranging (LiDAR) image, with 349 × 1950 pixels and a spatial resolution of 2.5 m. The HS image has 144 spectral bands covering a wavelength range from 0.364 to 1.046 μm. The LiDAR image provides height information in a single band. Field surveys were conducted to annotate 15 land use and land cover (LULC)-related classes. Table II presents the category names and information on the training and test sample sizes. Fig. 6 provides a visual representation and details on the training set and testing set can be found in Table II.

B. Experimental Settings

1) Assessment Metrics: To evaluate the effectiveness of the multimodal RS land cover classification, we employ three widely used assessment metrics: overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ). In general, those feature learning-based models consist of multimodal feature extractors followed by a classifier. This allows for a quantitative comparison with other competing approaches. More specifically, the above three metrics are given as follows:

$$\mathrm{OA} = \frac{\sum_{k=1}^{K} G_k \cap P_k}{N} \tag{11}$$

where G_k represents the actual number of class k labels, P_k stands for the predicted number of class k labels, K denotes the total number of categories, and N represents the overall number of samples across all categories

$$\mathrm{AA} = \frac{1}{K} \sum_{k=1}^{K} \frac{G_k \cap P_k}{N_k} \tag{12}$$

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TABLE III
QUANTITATIVE COMPARISON OF SEVERAL METHODS USING THE BERLIN DATASET. BOLD FONT DENOTES THE BEST RESULTS

Class	Specialized Model		Generalized Model				
	FusAtNet [14]	S^2ENet [54]	MDL-Middle [60]	S2FL [62]	CCR [61]	MFT [19]	Our TeACFNet
Forest(C1)	76.12	77.99	72.37	83.3	**85.93**	69.65	81.36
Residential Area(C2)	80.51	72.41	73.68	57.39	68.07	67.46	**80.60**
Industrial Area(C3)	36.80	55.61	31.39	48.53	53.17	57.33	**66.90**
Low Plants(C4)	76.88	86.14	87.54	77.16	82.62	**88.56**	80.76
Soil(C5)	53.30	79.88	**85.86**	83.84	85.10	79.61	81.93
Allotment(C6)	19.99	69.68	66.87	57.05	63.02	**78.01**	59.73
Commercial Area(C7)	21.37	32.00	**41.46**	31.02	29.23	30.51	27.59
Water(C8)	60.83	80.71	79.38	61.57	68.78	75.85	**79.40**
OA (%)	68.52	72.30	72.18	62.23	69.85	68.90	**76.77**
AA (%)	55.17	69.30	67.32	62.48	66.99	68.38	**69.78**
κ	0.5416	0.6034	0.5985	0.4877	0.5716	0.5671	**0.6542**

where N_k refers to the total number of samples in category k

$$\kappa = \frac{\mathrm{OA} - p_e}{1 - p_e} \tag{13}$$

where p_e is defined in [59] as the hypothetical probability of chance agreement and can be calculated by

$$p_e = \frac{\sum_{k=1}^{K} G_k \times P_k}{N \times N}. \tag{14}$$

2) Comparison With State-of-the-Art Method for Multimodal RS Data: To demonstrate the effectiveness of our proposed TeACFNet, we conducted a comprehensive evaluation by comparing it with various cutting-edge multimodal fusion classification methods. Our evaluation included specialized models that are designed for specific modalities (e.g., HS and LiDAR), including S^2ENet [54] and FusAtNet [14]), as well as four generalized models for all modalities, which are the multimodal deep learning (MDL) model [60], the cross-channel reconstruction (CCR) model [61], the shared and specific feature learning (S2FL) model [62], and the multimodal fusion transformer (MFT) [19].

3) Implementation Details: Our model is implemented on the Pytorch platform, and all experiments are conducted on 2 NVIDIA RTX 2080Ti. The original input images are cropped into nonoverlapping patches of 11×11 pixels during the training and testing process. The detailed backbone architecture is shown in Fig. 2, and the proposed model is trained in two stages. In the first stage, the epoch is set to 100, and the initial learning rate (LR) is 0.001. The epoch in the second stage is set to 64, and the initial LR is 0.001. The LR is modified using a cosine annealing method. Throughout the training phase, Adam acts as an optimizer, changing the gradient of the model. The batch sizes of Berlin dataset and Houston2013 dataset are both set to 64. It should be noted that in mini-batches, all images in a class may have the same confounders unless we can sample a large number of images in each class to obtain sufficiently diverse samples. To address this problem, we construct an inaccurate ClEF estimator using the bias model in LfF [57] and select a mini-batch with a relative balance of confounders based on its inverse probability. By employing this approach, there is no need to sample excessively large batches for learning (3).

C. Results and Analysis on Berlin Dataset

The quantitative comparison of classification results for various multimodal methods is shown in Table III.

We can draw the following conclusions.

1) Compared with other multimodal RS data classification methods, our TeACFNet has significant improvement and achieves the state-of-the-art performance in OA, AA, and κ.
2) In our horizontal comparison of the increments of TeACFNet with respect to other methods, the increments of OA and κ are relatively significant among the three evaluation metrics. For example, TeACFNet achieves 11.42%, 2.05%, and 15.36% increments on OA, AA, and κ, respectively, compared with MFT (transformer-based generalized model), and other methods show consistent results.
3) For the classification accuracy of each class, our model obtains a significant improvement over MDL (best in the generalized models) and achieves better than other methods in classes "Residential Area," "Industrial Area," and "Water." For "Residential Area," our TeACFNet achieves significant gains over other methods, which shows that the two-stage based causal feature representation has a significant advantage in learning the category invariance of land cover.

Qualitative comparisons on the Berlin dataset are shown in Fig. 7. The test set in this dataset represents a small portion of the overall dataset, and the number of samples for each category varies significantly. As illustrated in Fig. 7, our proposed model exhibits significant improvements in the consistency of land cover classification results compared to other comparative methods, which exhibits considerable texture and color variations within the category. For instance, when classifying the "Residential Area" category, pixel-by-pixel classification networks, such as FusAtNet and MFT, tend to misclassify regions, whereas our model achieves accurate predictions. These experimental results demonstrate the effectiveness of our TeACFNet in capturing fundamental representation differences between categories and emphasize the significance of CFE in multimodal RS data classification, addressing previous unresolved challenges. In general,

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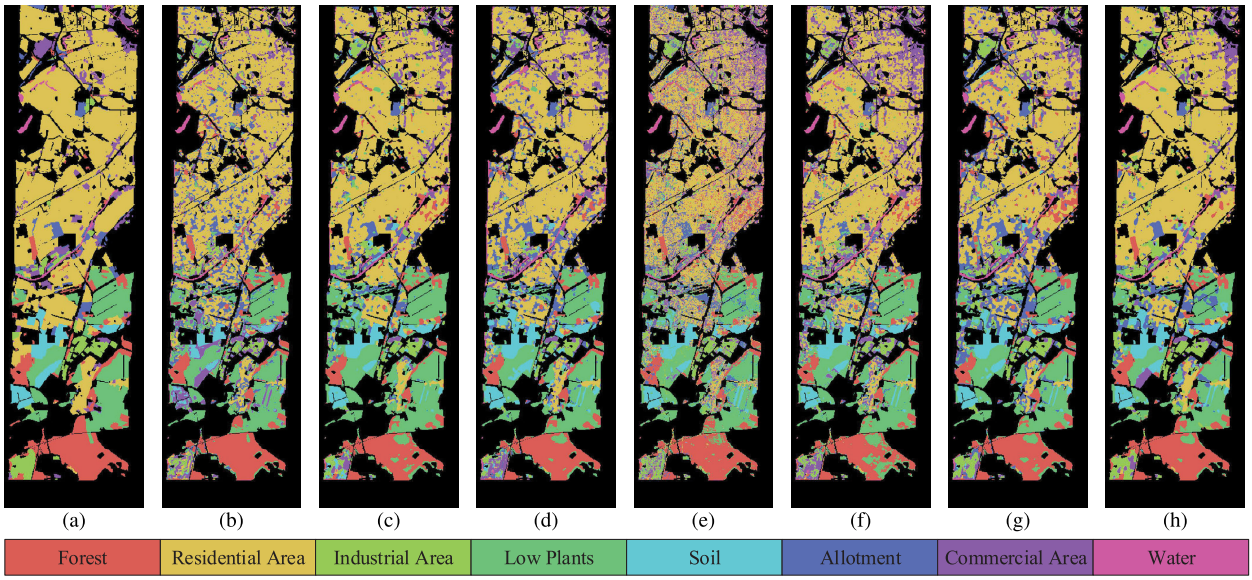


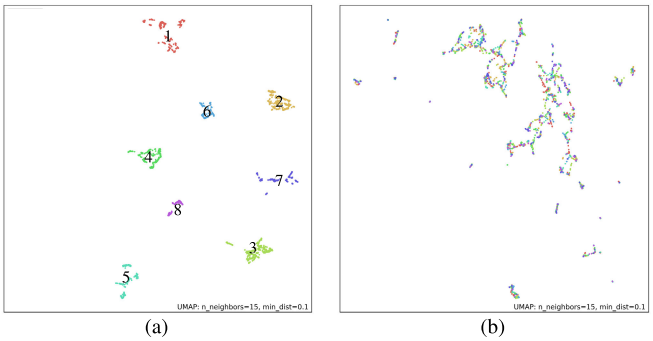
Fig. 7. Classification maps for Berlin dataset. (a) GT. (b) FusAtNet. (c) S^2ENet. (d) MDL-Middle. (e) S2FL. (f) CCR. (g) MFT. (h) Our TeACFNet.

TABLE IV
QUANTITATIVE COMPARISON OF SEVERAL METHODS USING THE HOUSTON2013 DATASET. BOLD FONT DENOTES THE BEST RESULTS

Class	Specialized Model		Generalized Model				
	FusAtNet [14]	S^2ENet [54]	MDL-Middle [60]	S2FL [62]	CCR [61]	MFT [19]	Our TeACFNet
Healthy grass(C1)	83.10	82.91	83.10	80.06	83.00	82.34	82.15
Stressed grass(C2)	96.05	100	85.06	84.49	84.81	88.78	99.72
Synthetic grass(C3)	100	100	99.60	98.02	100	98.15	99.80
Trees(C4)	93.09	96.88	91.57	87.31	92.14	94.35	96.69
Soil(C5)	99.43	99.91	98.86	100	99.81	99.12	98.96
Water(C6)	100	100	100	83.22	95.80	99.30	100
Residential(C7)	93.53	95.15	96.64	73.32	95.34	88.56	95.34
Commercial(C8)	92.12	93.92	88.13	74.84	81.39	86.89	92.97
Road(C9)	83.63	91.31	85.93	78.38	84.14	87.91	89.05
Highway(C10)	64.09	92.95	74.42	83.3	63.22	64.70	97.78
Railway(C11)	90.13	94.69	84.54	81.69	90.32	98.64	96.77
Parking Lot 1(C12)	91.93	89.43	95.39	95.10	93.08	94.24	89.24
Parking Lot 2(C13)	88.42	83.16	87.37	72.63	88.42	90.29	82.46
Tennis Court(C14)	100	100	95.14	100	96.36	99.73	98.79
Running Track(C15)	100	100	100	99.37	99.37	99.58	99.79
OA (%)	89.92	94.19	89.55	85.07	88.15	89.80	**94.25**
AA (%)	**94.65**	94.69	91.05	86.11	89.82	91.51	94.63
κ	0.8913	0.9369	0.8871	0.8378	0.8719	0.8893	**0.9376**

the proposed TeACFNet is able to produce more accurate pixel-level classification maps, especially for complex irregular objects, and obtain finer boundary details.

In our UMAP [63] visualizations, as shown in Fig. 8, causal features display distinct category-based clustering, highlighting their direct relevance to class definitions. Conversely, ClEFs show a more scattered distribution, lacking clear category association, which aligns with our hypothesis that these features do not primarily drive classification.



(a) (b)

Fig. 8. Visualization of causal features and ClEFs of Berlin dataset obtained by UMAP: (1)–(8) are labels that can be obtained from Table I. (a) Causal features extracted by the extractor φ_c. (b) ClEFs extracted by the extractor φ_s.

D. Results and Analysis on Houston2013 Dataset

Similarly, Table IV provides a detailed quantitative comparison of various multimodal methods applied to the HS-LiDAR Houston2013 dataset. Furthermore, Fig. 9 visually illustrates the corresponding classification maps obtained from these algorithms. We observe a similar trend in classification performance as observed in the HS-SAR Berlin dataset.

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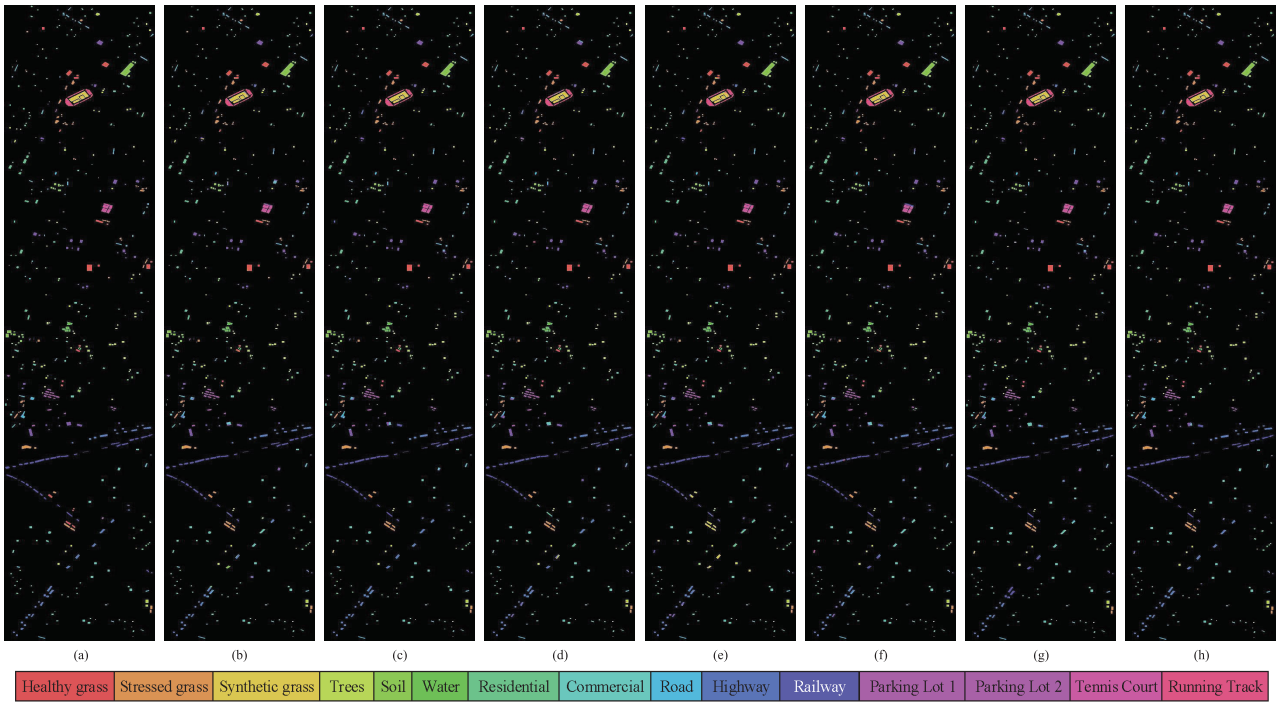


Fig. 9. Classification maps for Houston2013 dataset. (a) GT. (b) FusAtNet. (c) S^2ENet. (d) MDL-Middle. (e) S2FL. (f) CCR. (g) MFT. (h) Our TeACFNet.

Specifically, the integration of the refined statistical texture feature extractor enhances the deep-level perception of texture details within the backbone network, while the two-stage CFE framework effectively filters out spurious features, allowing for better land cover identification.

Our proposed TeACFNet exhibits significantly better performance than all other multimodal RS classification methods. Notably, our model achieves an OA of 94.25%, AA of 94.63%, and a kappa coefficient of 0.9376. Although FusAtNet achieves a slightly higher AA, the marginal difference of 0.02% is negligible compared to the significant success of our model in generalization. By visually analyzing the classification maps obtained from our proposed method, we observe that the noise is significantly reduced and smooth transitions between categories are achieved. This improvement can be attributed to the effective utilization of multimodal data through the two-stage CFE process.

TABLE V

ABLATION EXPERIMENT RESULTS OF THE PROPOSED TeACFNet. ✓MEANS ADD MODULE

Component	Berlin dataset			
ClEF+I^2FE	×	✓	×	✓
ReSTE	×	×	✓	✓
OA(%)	71.93	74.99	73.86	76.77
AA(%)	64.51	67.77	63.77	69.78
κ	0.5824	0.6241	0.6100	0.6542

TABLE VI

OPERATIONAL EFFICIENCY AND MODEL COMPLEXITY OF SEVERAL METHODS ON THE BERLIN DATASET

Model name	AIT(s)	Total Params	FLOPs
MDL-Middle [60]	90.565	133.545K	7.133M
MFT [19]	95.893	394.409K	43.422M
S^2ENet [54]	95.682	386.090K	46.732M
FusAtNet [14]	504.154	3.562G	37.719M
Our method	97.859	532.073K	48.755M

E. Ablation Study

To investigate the impact of different components in our proposed model on classification performance, we conducted a series of experiments by systematically removing specific parts of the model. This ablation study was carried out on the Berlin dataset to gain insights into the individual contributions of these components. The experimental results are summarized in Tables V and VI and Figs. 10 and 11.

1) With/Without the ClEF and I^2FE Module: In assessing the individual contributions of the ClEF and I^2FE modules to our TeACFNet, we conducted an ablation study. These modules serve distinct purposes: ClEF focuses on learning ClEFs that remain consistent despite changes in the environment, i.e., categories. I^2FE aims to extract causal features by utilizing IPW, thus mitigating context bias in land cover classification. The results of this ablation study are summarized in Table V. It is evident that removing these two modules leads to a significant decrease in the accuracy of land cover classification. The main reason is that when these modules are removed, the causal features in the network are no longer effectively decoupled, and the quality of feature maps decreases. Fig. 10 demonstrates that our proposed TeACFNet can identify more precise category region boundaries, which further proves this point.

2) With/Without the ReSTE Module: The experimental results without the ReSTE module are presented in Table V. From the table, it is evident that the removal of the ReSTE

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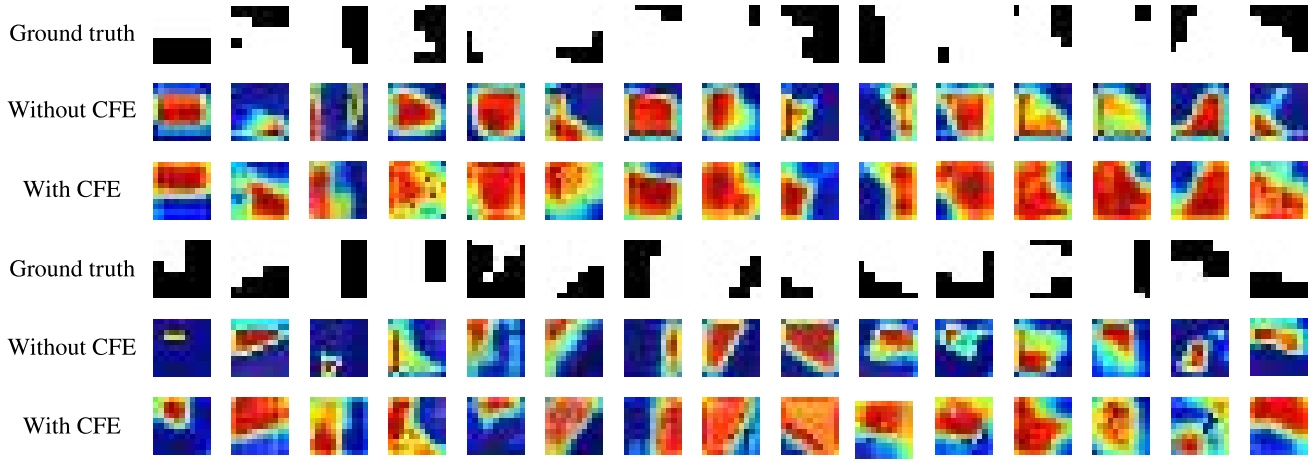
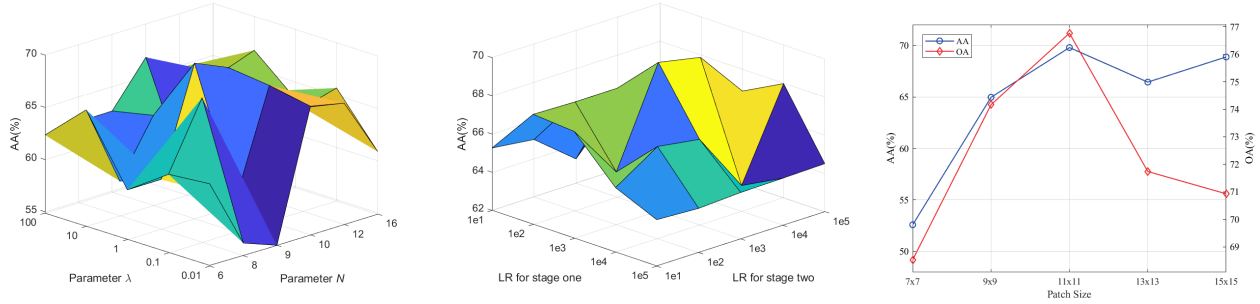


Fig. 10. GradCAM [64] visualizes the learned causal feature heatmap and compares it with the feature heatmap obtained without using the two-stage CFE. The white part in the GT represents the foreground and the black part represents the background.



(a) (b) (c)

Fig. 11. Analysis of hyperparameters on Berlin dataset. (a) Hyperparameter λ and N. (b) Hyperparameter LR for stages 1 and 2. (c) Hyperparameter patch size.

module has a substantial negative impact on classification accuracy. The results demonstrate that the inclusion of the ReSTE module enhances the ability of the model to learn and capture intricate textural features and spatial patterns. Fig. 4 shows the visualization comparison on feature maps before ReSTE and after ReSTE. The refined output consistently exhibits higher activation values in the concrete building pixel locations, highlighting the module's ability to emphasize relevant textural details associated with the industrial area. This indicates that the ReSTE module effectively captures and emphasizes relevant texture details of objects of the same category, thus improving feature representation and discrimination.

3) Study of Hyperparameters: In the proposed TeACFNet, there are five hyperparameters that directly affect the performance of the network, including the hyperparameter λ that balances the IRM weights, the quantized number N of texture feature maps, LR for stages 1 and 2, and the size of the patch size. λ serves as a tradeoff parameter for the invariance regularization term. The quantized number N of texture features controls the representation ability of statistical texture features. The ablation research on the hyperparameter λ and quantization number N on the Berlin dataset is shown in Fig. 11(a). Obviously, the larger N is, the higher the computational complexity of the model is. When $N = 9$ and $\lambda = 1.0$, the model achieves the best OA of 76.77%. The effects of LR for two stages and the size of patch size on the classification accuracy of the model were tested, as shown in Fig. 11(b) and (c), respectively.

4) Study of Operational Efficiency and Model Complexity: We calculate the average inference time (AIT), total parameters, and floating point operations per second (FLOPs) of several state-of-the-art models on the Berlin dataset. Table VI presents the results obtained from the experiment. Analyzing the table, we can observe that our method shows a modest rise in AIT, yet stays well within practical limits. In terms of model complexity, our approach strikes a balance. It possesses a moderate number of total parameters while having higher FLOPs compared to other models.

Moreover, our approach demonstrates a remarkable tradeoff between model complexity and performance. It achieves competitive results in land cover classification while efficiently utilizing computational resources. This underscores our method's effectiveness in achieving accurate classification outcomes while maintaining operational efficiency. By optimizing the utilization of computational resources and maintaining a reasonable model complexity, our approach is scalable and ready for real-world deployment.

V. Conclusion

In this article, we introduce IRM into multimodal RS data for the first time and propose the novel TeACFNet for RS land cover classification. Our experimental results demonstrate the remarkable performance of our proposed model compared to other models. The proposed ClEF and I^2EF play a pivotal role in extracting features causally linked to classification tasks, leading to significant enhancements in

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TeACFNet's performance. By utilizing IRM on the intraclass sample similarity contrastive loss, the ClEF ensures context invariance, while the use of IPW enables CFE without the need for additional attribute annotations. Furthermore, the ReSTE module captures essential statistical texture details by analyzing local features and their interconnections within shallow feature maps, substantially boosting the discriminative capabilities of the feature extractor and resulting in improved land cover classification accuracy.

The current model achieves excellent performance in multimodal land cover classification. However, existing multimodal land cover classification datasets are usually multimodal images aligned. In future work, we prefer to combine unsupervised domain adaptation to further study cross-scene and cross-modal land cover classification problems.

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